

Networks and Security

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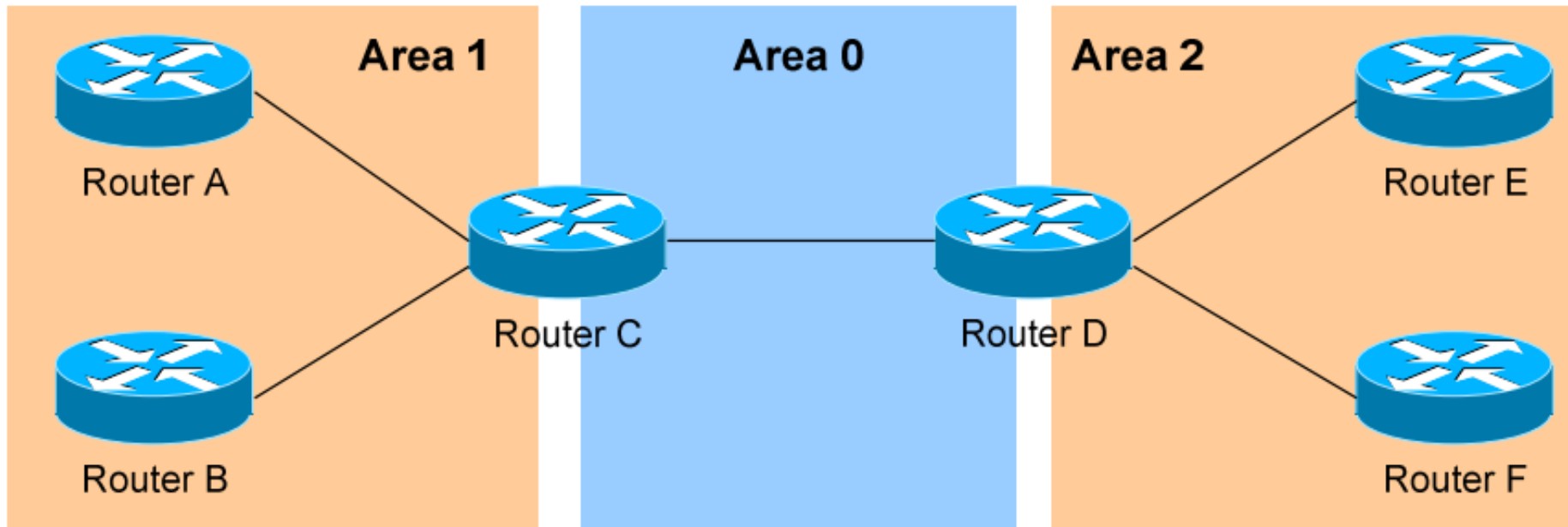
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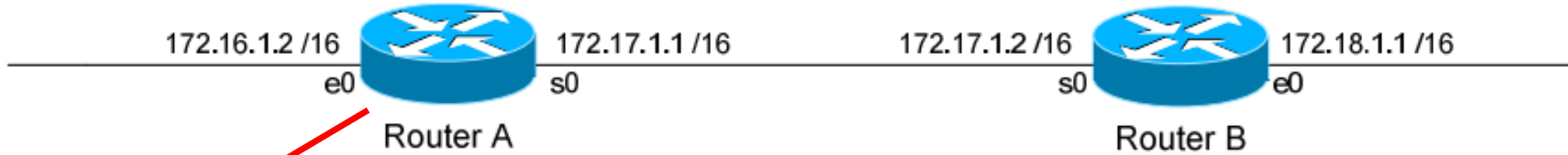
Open Shortest Path First (OSPF)

The OSPF Hierarchy

- OSPF is a hierarchical system that separates an Autonomous System into individual areas.



Configuring Basic OSPF



RouterA(config)# router ospf 1

RouterA(config-router)# network 172.16.0.0 0.0.255.255 area 0

RouterA(config-router)# network 172.17.0.0 0.0.255.255 area 0

RouterB(config)# router ospf 2

RouterB(config-router)# network 172.18.0.0 0.0.255.255 area 0

RouterB(config-router)# network 172.17.0.0 0.0.255.255 area 0

OSPF command

R1(config)#router ospf 1

Router(config-router)#network network-address wildcard-mask area area-id

- The inverse of the subnet mask results in the wildcard mask.
- To calculate the wildcard mask:

$$\begin{array}{r} 255.255.255.255 \\ - 255.255.255.252 \text{ (Subtract the subnet mask)} \\ \hline \end{array}$$

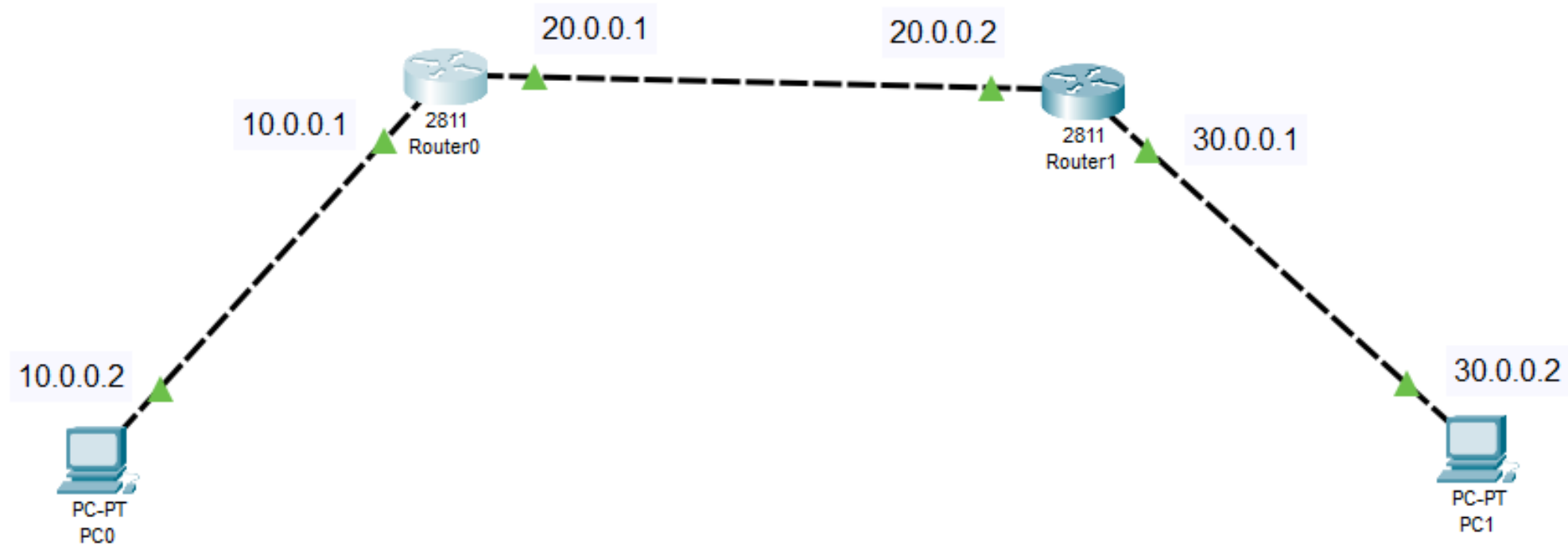
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Wildcard mask

Exercise

Objectives

Design the given topology and configure OSPF to demonstrate that the PCs are successfully connected.



Configure IP addresses

R1(config)#int fa 0/0

R1(config-if)#ip add 10.0.0.1 255.0.0.0

R1(config-if)#no shut

R1(config-if)#

R1(config-if)#int serial 0/0/0

R1(config-if)#ip add 20.0.0.1 255.0.0.0

R1(config-if)#no shut

R2(config-if)#int fa0/0

R2(config-if)#ip add 30.0.0.1 255.0.0.0

R2(config-if)#no shut

R2(config-if)#

R2(config-if)#int serial0/0/0

R2(config-if)#ip address 20.0.0.2 255.0.0.0

R2(config-if)#no shut

PC1 IP add 10.0.0.2 Subnet mask 255.0.0.0 Default gateway 10.0.0.1

PC2 IP add 30.0.0.2 Subnet mask 255.0.0.0 Default gateway 30.0.0.1

Configure OSPF

R1(config)#

R1(config)#router ospf 1

R1(config-router)#network 10.0.0.0 0.255.255.255 area 0

R1(config-router)#network 20.0.0.0 0.255.255.255 area 0

R2(config)#

R2(config)#router ospf 2

R2(config-router)#network 20.0.0.0 0.255.255.255 area 0

R2(config-router)#network 30.0.0.0 0.255.255.255 area 0

Verify OSPF configuration & connectivity between PCs

R1# show ip ospf neighbor

R1#show ip route ospf

R2# show ip ospf neighbor

R2#show ip route ospf

Ping PC1 from PC2